

t -Distribution

CSU, Chico Math 351

2026-03-09

outline

From Normal to T-Test
Confidence Interval
Hypothesis Test

Bernoulli

Normal

References

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t , confidence intervals

Confidence intervals under the t -distribution are nearly identical,

$$\bar{X} \pm t_{1-\alpha/2, df}^* \cdot s_{\bar{X}}.$$

t , hypothesis tests

Hypothesis tests under the t -distribution are nearly identical, standardizing the sample statistic as before:

$$t_{df} = \frac{\bar{X} - \mu_0}{s_{\bar{X}}}.$$

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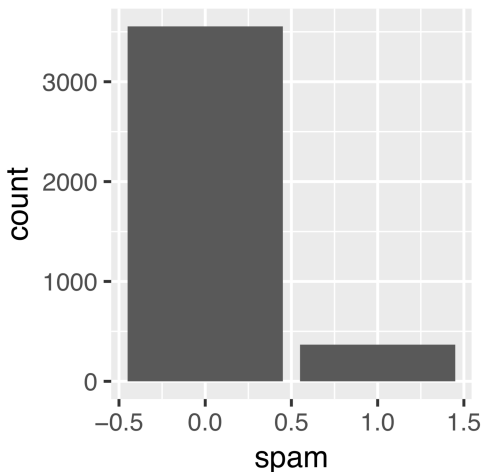
References

Proportion of emails spam

Consider the following `email` data set. Calculate and interpret a 98% confidence interval for the proportion of emails that are spam.

Proportion of emails spam

```
email <- read.csv("https://raw.githubusercontent.com/  
roualdes/data/master/email.csv")  
ggplot(data=email, aes(spam)) + geom_bar()
```



Proportion of emails spam

```
phat <- mean(email$spam)
n <- length(email$spam)
s <- sd(email$spam)
alpha <- 0.02

se <- s/sqrt(n)
t.star <- qt(1-alpha/2, df=n-1)
ME <- se * t.star

c(phat - ME , phat + ME)    # confidence interval (CI)

## [1] 0.08277167 0.10442547
```

Proportion of emails spam, conclusion

We are 98% confident that the true proportion of emails that are spam is between 0.0828 and 0.104.

Proportion of emails spam, extra

Follow up questions.

1. What sort of issues with these data should we keep in mind?
2. Does the interval above include the true population proportion of interest?
3. Would a 90% confidence interval be wider or narrower? 99%?
4. How could you use this interval to conclude the following hypothesis test?

$$H_0 : p = 0.15 \text{ versus } H_1 : p \neq 0.15; \alpha = 0.02$$

5. What would be the 98% confidence interval for the proportion of all emails that are HTML?

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Normal

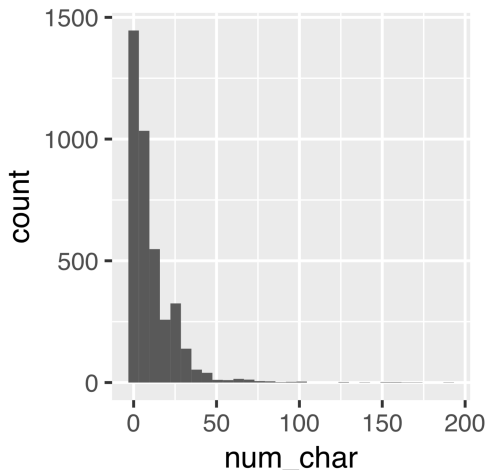
References

Mean number of characters per email

Consider the `email` data set. Calculate and interpret a 95% confidence interval for the mean number of characters per email.

Mean number of characters per email

```
ggplot(data=email, aes(num_char)) +  
  geom_histogram(bins=31)
```



Mean number of characters per email

```
xbar <- mean(email$num_char)
s <- sd(email$num_char)
n <- length(email$num_char)
alpha <- 0.05

se <- s/sqrt(n)
t.star <- qt(1-alpha/2, df=n-1)
ME <- se * t.star

c(xbar - ME , xbar + ME)    # confidence interval (CI)

## [1] 10.24803 11.16515
```

Mean number of characters per email, extra

We are 95% confident that the true mean number of line characters per email is between 10.25 and 11.17.

Mean number of characters per email, extra

Follow up questions.

1. What sort of issues with these data should we keep in mind?
2. Does the interval above include the true population mean of interest?
3. Does more confidence mean a wider interval or smaller?
4. How could you use this interval to conclude the following hypothesis test?

$$H_0 : \mu = 11 \text{ versus } H_1 : \mu \neq 11; \alpha = 0.05$$

5. What would be the 95% confidence interval for the mean number of line breaks in all emails?

An Additional Problem

A lightbulb manufacturer makes the claim that its fluorescent lightbulbs last for 12,000 hours. After testing 64 such lightbulbs, you find that the average lifespan of the lightbulbs is 11,297 hours, with a sample standard deviation of 2,418 hours. Construct both a 95% confidence interval and an appropriate hypothesis test with $\alpha = 0.05$ to analyze the manufacturer's claim.

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references I

David M Diez, Christopher D Barr, and Mine Cetinkaya-Rundel. *OpenIntro Statistics*. CreateSpace independent publishing platform, third edition, 2015.

Hadley Wickham. *ggplot2: elegant graphics for data analysis*. Springer Science and Business Media, 2009.

Adapted from notes by Dr. Roualdes.